LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

Ubuntu Server Boot Process Flowchart

ITEP 414 | Week 3

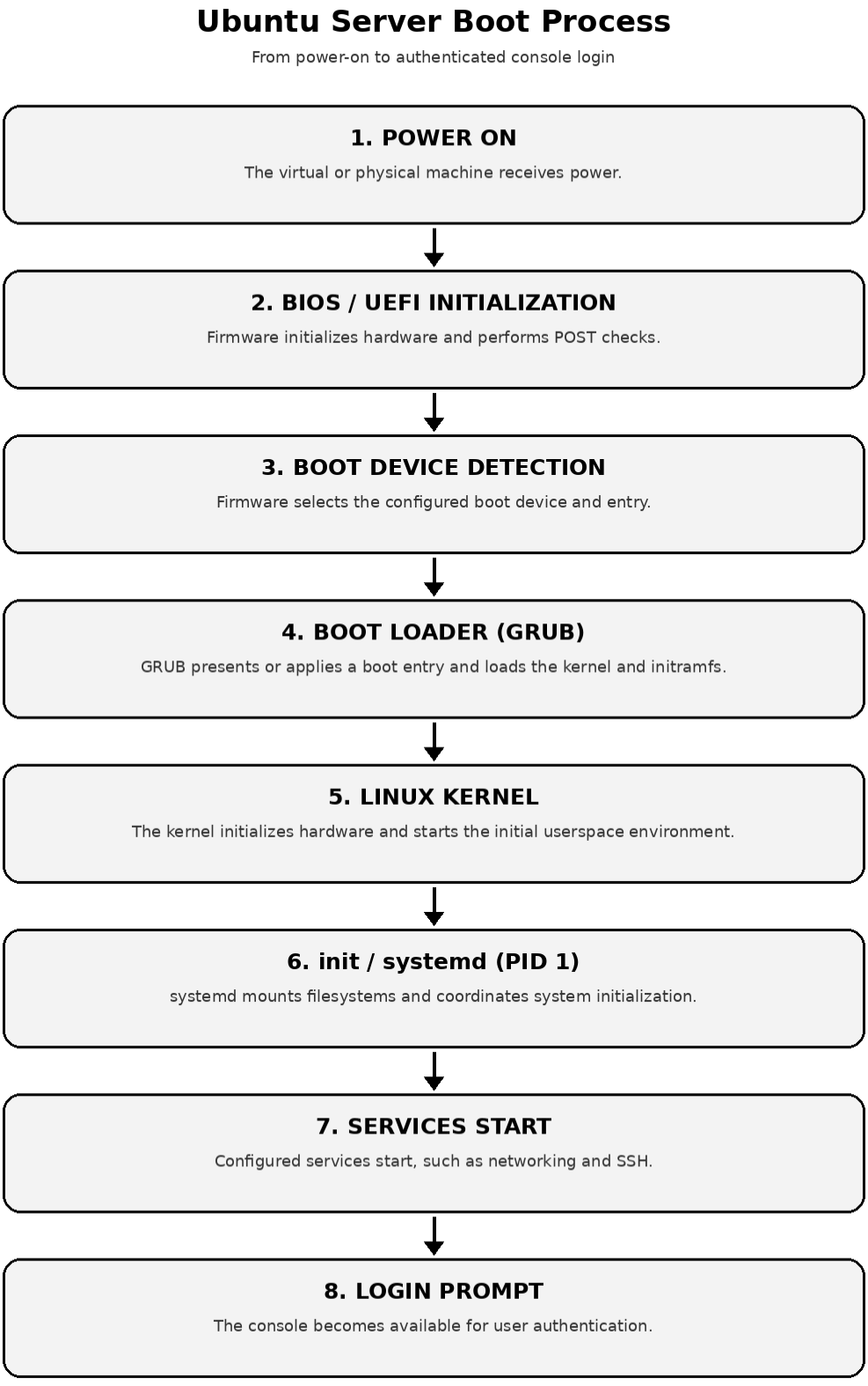
A high-level sequence showing how the system moves from firmware initialization to a usable login prompt.

Project Information	Details
System	Ubuntu Server
Boot loader	GRUB (as specified in the project brief)
Init system	systemd on a typical modern Ubuntu Server installation

Prepared for ITEP 414 - System Administration and Maintenance

Working template: replace bracketed fields and add genuine lab evidence before submission.

Flowchart



Note: exact screens and service timing depend on firmware, VM settings, and installed services.

How to Read the Flowchart

Firmware performs early initialization and identifies a bootable device. The boot loader loads the Linux kernel and initial RAM filesystem. The kernel initializes hardware and transfers control to the userspace init system. systemd coordinates startup targets and services; when the configured console is ready, the login prompt is available.

This is a simplified learning diagram. Actual startup details vary with firmware mode, GRUB configuration, kernel parameters, storage layout, and enabled services.

Required Sequence

Step	Stage	Purpose
1	Power On	Begins the startup sequence.
2	BIOS / UEFI Initialization	Initializes platform hardware and performs startup checks.
3	Boot Device Detection	Chooses the configured boot device or firmware boot entry.
4	Boot Loader (GRUB)	Loads the selected kernel and initramfs.
5	Linux Kernel	Initializes the operating system core and prepares userspace.
6	init / systemd	Coordinates system initialization and dependency ordering.
7	Services Start	Starts configured services, including networking and SSH if enabled.
8	Login Prompt	Allows a user to authenticate at the console.

Reference

Ubuntu Server installation documentation: https://ubuntu.com/server/docs/installation/

systemd manual: https://www.freedesktop.org/software/systemd/man/latest/systemd.html

GNU GRUB manual: https://www.gnu.org/software/grub/manual/grub/grub.html